

YUNCHUAN LIU

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<https://scholar.google.com/citations?user=0bbF8hAAAAAJhl=en>

EDUCATION

University of Nevada,Reno

August 2017 - August 2022

Doctor of Philosophy in Computer Science and Engineering.

Theses: Towards Automated Machine Learning on Imperfect Data for Situational Awareness in Power System

Shenzhen University, Guangdong, China

September 2009 - June 2012

Master Degree of Science in Optics.

Theses: Circuit design of micro projector video processing system based on FPGA

Shenzhen University, Guangdong, China

September 2003 - July 2007

Bachelor Degree of Science in Optical Information.

Theses: Thin film design of high reflection in infrared wave

WORK EXPERIENCE

-Governors State University 2022.8 - now Assistant Professor(tenure track)

-University of Nevada, Reno 2017.8 - 2022.8

Research Assistant: focusing on big data and machine learning related applications

INTEREST

Data Analytics and Stochastic Optimization for the Integration of Renewable Energy, Data Mining and Data Engineering, Smart Grids, Embedded system.

TEACHING

1. CPSC-3310 Intro To Object-oriented Programming
2. CPSC-6548 Computer Programming: Java
3. CPSC-6780 Big Data Processing and Analytics
4. CPSC-4792\6792: Language Model And Processing
5. CPSC-8810 Formal Languages and Automata
6. CPSC-8845 Advanced Database Concepts
7. CPSC-6790 Data Mining and Business Intelligence
8. CPSC-8985 Grad Seminar

FUNDING

- 1) 2026-2027: Conduct an analysis and recommendations for University preparedness, planning, and mitigation for energy and water impacts of data center projects across the IIN (Sole-PI) is awarded by Illinois Innovation Network Fellow Program \$15k .
- 2) 2024-2025: MediValAir: Innovative Drone Delivery System for Hospitals(Co-PI) is awarded by Illinois Innovation Voucher \$75k.

CONFERENCE PAPERS

- 1) Mahmoud, Yunchuan Liu,Abraham Canafe and Lei Yang, Towards distributed learning of PMU data: A federated learning based event classification approach. 2023 IEEE Power & Energy Society General Meeting (PESGM). IEEE, 2023.

- 2) Yunchuan Liu and Lei Yang, Weakly supervised event classification using imperfect real-world PMU data with scarce labels. 2022 IEEE Power & Energy Society General Meeting (PESGM). IEEE, 2022. (Best Paper Award)
- 3) Amir Ghasemkhani, Yunchuan Liu and Lei Yang, Optimized Event Detection Framework for Synchrophasor Data. 2022 IEEE Power & Energy Society General Meeting (PESGM). IEEE, 2022.
- 4) Abraham Canafe, Yunchuan Liu, Lei Yang, Hanif Livani, DCCA Enhanced Forced Oscillation Frequency Detection Using Real-world PMU Data. In 2022 IEEE Texas Power and Energy Conference (TPEC) (pp. 1-6). IEEE. (REU project to mentor undergrad student work)
- 5) Yunchuan Liu, Amir Ghasemkhani, Lei Yang, Jun Zhao, Junshan Zhang, and Vijay Vittal, Seasonal self-evolving neural networks based short-term wind farm generation forecast. In 2020 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm) (pp. 1-6). IEEE.
- 6) Iman Niazazari, Yunchuan Liu, Amir Ghasemkhani, Shuchismita Biswas, Hanif Livani, Lei Yang, Virgilio A. Centeno. (2021, February). Pmu-data-driven event classification in power transmission grids. In 2021 IEEE Power Energy Society Innovative Smart Grid Technologies Conference (ISGT) (pp. 1-5). IEEE.
- 7) Iman Niazazari, Hanif Livani, Amir Ghasemkhani, Yunchuan Liu, Lei Yang. (2021, April). Event cause analysis in distribution networks using synchro waveform measurements. In 2020 52nd North American Power Symposium (NAPS) (pp. 1-5). IEEE.
- 8) Amir Ghasemkhani, Yunchuan Liu, Lei Yang. (2021, February). Low-rank Tensor Completion for PMU Data Recovery. In 2021 IEEE Power Energy Society Innovative Smart Grid Technologies Conference (ISGT) (pp. 1-5). IEEE.
- 9) Yunchuan Liu, Shuang Li, and Zeyang Xia. (2015, December). Path planning efficiency maximization for ball-picking robot using machine learning algorithm. In 2015 International Conference on Intelligent Transportation, Big Data and Smart City (pp. 551-555). IEEE.
- 10) Yunchuan Liu, Yingdong Huo and Hanben Niu. (2015, December). A method for reducing the sidelobes in superoscillation imaging. In MIPPR 2015: Multispectral Image Acquisition, Processing, and Analysis (Vol. 9811, pp. 76-81). SPIE.

JOURNAL PAPERS

- 1) Liu, Yunchuan, Lei Yang, and Junshan Zhang. Automatic Labeling of Real-World PMU Data: A Weakly Supervised Learning Approach. *Electronics* 14.23 (2025): 4703.
- 2) Haoyan Cheng, Mingyong Song, and Yunchuan Liu. Related Technological Density and Regional industrial Upgrading from Perspective of Product Space Theory: Evidence from China. *Applied Economics* (2024): 1-15.
- 3) Yunchuan Liu, Amir Ghasemkhani and Lei Yang. Drifting Streaming Peaks-over-Threshold Enhanced Self-evolving Neural Networks for Short-term Wind Farm Generation Forecast, *Future Internet* 15.1 (2023): 17.
- 4) Yunchuan Liu, Lei Yang, Amir Ghasemkhani, Hanif Livani, Virgilio A. Centeno, Pin-Yu Chen, Junshan Zhang, J. (2022). Robust Event Classification Using Imperfect Real-world PMU Data. *IEEE Internet of Things Journal*.
- 5) Amir Ghasemkhani, Iman Niazazari, Yunchuan Liu, Hanif Livani, Virgilio A. Centeno, Lei Yang. (2020). A regularized tensor completion approach for pmu data recovery. *IEEE Transactions on Smart Grid*, 12(2), 1519-1528.
- 6) Yunchuan Liu, Heng Li, Danni Chen, Hanben Niu. (2016). A Combined SOM/SVM Learning Algorithm for Vibration Recognition in Nanometer Imaging Systems. *International Journal of Simulation-Systems, Science & Technology*, 17(36).
- 7) Yunchuan Liu, Junshan Yang and Hanben Niu. (2015). A Study on Vibration Recognition of Nano-imaging System Based on Wavelet Analysis. *The Open Automation and Control Systems Journal*, 7(1).
- 8) Yunchuan Liu and Xiangdong Gong. (2013). Processing and Hardware Implementation of BT. 656 Digital Video Stream. *Chinese Journal of Liquid Crystals and Displays*, 28(2), 238-243.
- 9) Zuolin Cheng, Yunchuan Liu and Xiangdong Gong. (2012). Design of USB Transmission System for Micro projection Video Signal. *Chinese Journal of Liquid Crystals and Display*, 27(1), 81-86.

- 10) Yunchuan Liu, Xiangdong Gong, Wu Qingyang. (2011). SPI IP Core and Its Application in Microprojection System. Microcontroller and Embedded Systems, (2), 27-30.

SERVICES

1. NSF reviewer (2025,2026).
2. Reviewer for GSU Scholarships(2026).
3. Chair in GSU Computer Science Program Curriculum Assessment Committee (2022-now).
4. Guest editor for 'Electronics' journal(JCR Q2) under MDPI (2025-2026).
5. Senate in GSU Graduate Council (2023-2025).
6. Committee member in GSU Computer Science Program Search Committee (2022-2025).
7. Committee member in GSU Division Curriculum Committee (2022-2024).
8. Review of manuscripts for academic journals and serving on conference panels. I am the reviewer for IEEE Transactions on Wireless Communications, IEEE Internet of Things, IEEE Access and Journals under Multidisciplinary Digital Publishing Institution (MDPI).
9. Serve as a Referee at 2023 IEEE Power & Energy Society General Meeting (PESGM) and 2023-2025 Illinois Louis Stokes Alliance for Minority Participation (ILSAMP).

AWARD

- Best Conference Paper Award in PES General Meeting 2022(top1%).
- 1st prize in China-US Young Maker Competition,CUYMC,Shenzhen Division(top1%) in 2017.
- 2nd prize in "Challenge Cup" Competition for University Students Guangdong Province (top3%) in 2011.
- 1st prize in 2010 Asian Innovation Altera FPGA Design Contest (top 1%) in 2010.